LAB MANUAL

CHY1005: Introduction to Computational Chemistry

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School of Advanced Sciences and Languages, Chemistry Introduction to Computational Chemistry

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Registration Number: 23BAI10917 **Duration: 90 min.**

Practical 2: Single Point Energy Calculation of Molecules

Aim

To calculate single point energy of molecules using a quantum mechanical method.

Theory:

Single point energy is the potential energy of a molecule for a given arrangement of the atoms in the molecule. Single point energies are the lowest energy solution for the Schrödinger equation and are the simplest properties one might aim to obtain. It consists in the calculation of the wave function and energy for a given system with a well-specified geometric structure, i.e. at a single, fixed point on the potential energy surface. The computed energy is the total energy, sum of the electronic energy and nuclear repulsion energy. To perform a single point energy calculation a well-defined level of calculation must be specified. A level of calculation is uniquely defined by the combination of a theoretical method with a basis set.

Many aspects of the system"s behaviour can be understood when the total energy of the system is known, or more often, when the energy difference between two or more electronic or nuclear configurations is known as a function of some parameters.

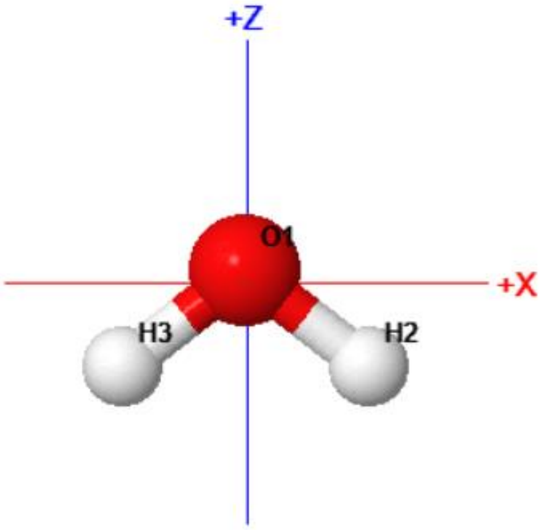
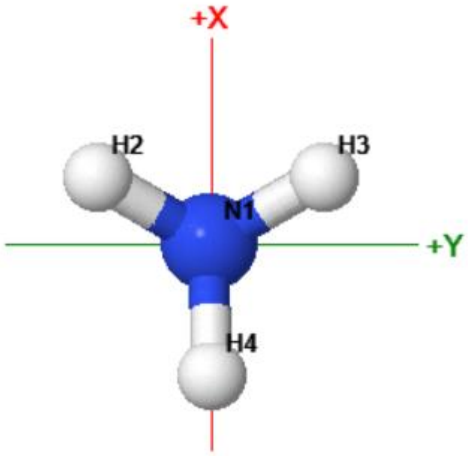
Procedure:

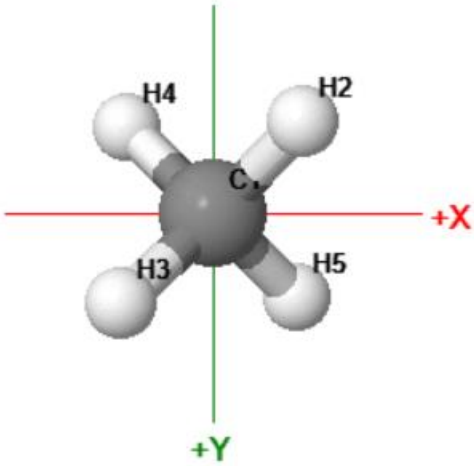
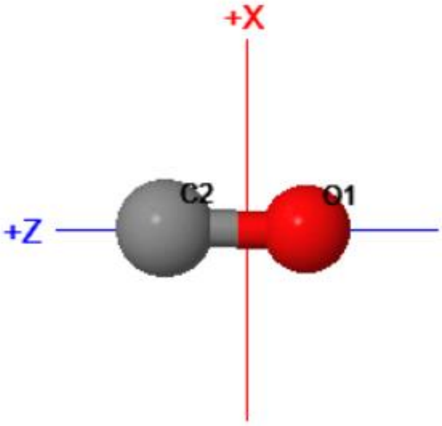
- Build a water molecule using ChemCompute server or any other molecular builder.
- Prepare the input for single point energy calculation. Choose an appropriate method and basis set.
- Submit the job by clicking on submit button.
- Note down the electronic energy of the water molecule.

- To perform calculations on a new molecule click the Submit Tab at the top.

Results:

Draw the following molecules and report their single point energy.

S. No.	Molecule	Single Point Energy (Hartree)
1	H_2O 	-12.8092313402
2	He_2	**Error/Failure in Loading**
3	NH_3 	-9.1343839360

4	CH_4 	-6.7322902652
5	CO 	-16.4257536336